LIBBY — PATIENT/CAREGIVER TRACK

cca-mets-scope-unknown-z3p9

Plain-language summary for patient and caregiver

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This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

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What this page is

This is the patient and caregiver version of a longer report Libby put together about treatment options for metastatic cholangiocarcinoma, a cancer of the bile ducts. The case Libby was handed had almost nothing in it: the diagnosis, and little else. No genetic testing of the tumor had been done yet.

That one fact shapes everything below. Cholangiocarcinoma can carry any of several specific genetic changes, and for most of those changes there is a drug built to go after it. But you cannot know which of those drugs might help until you know what the tumor is actually made of. So the page is organized around one honest reality: there is a single test that comes before any of the drug choices, and it is the test that tells you which door is open. The drug choices only come into focus once that test is back.

A couple of things to know going in. Libby only looks at treatments aimed at a specific genetic feature of the tumor. The ordinary chemotherapy that most people with this cancer start on is real and important, but it is not what this page covers, and your oncologist owns that conversation. And because no test results were available, a lot of what would normally personalize a plan (your other health conditions, what you care about most, what you have already tried) was filled in with neutral placeholders. Treat everything here as a starting point for a conversation, not a plan.

What we know about your cancer

Not much was provided, and it is worth being upfront about that.

The cancer started in the bile ducts (cholangiocarcinoma) and has spread beyond where it began (metastatic, stage IV). Whether it started inside the liver or in the ducts outside the liver was not recorded, and that detail matters, because some of the genetic changes are more common in one location than the other.

No genetic testing of the tumor had been done. Performance status was assumed to be relatively good (up and about, able to care for yourself), age was assumed to be in the 60s, and no prior treatments were on record. These are placeholders, not facts about you. If any of them is wrong, the plan should be revisited.

What you told us matters most

No preferences were recorded for this case, so Libby used a balanced setting: how well a treatment works and how hard its side effects hit were weighed equally, with nothing ruled in or out ahead of time. One thing was noted, that being open to clinical trials counts as a plus. If you have strong feelings, say a real fear of a particular side effect, or a preference for pills over infusions, those would shift the order of what follows, and they are worth telling your oncologist directly.

The first step everyone agreed on

Before any drug below is on the table, the tumor needs to be genetically tested. This is called comprehensive genomic profiling, and it is the clear first move. Think of it as the test that tells us which door is open, because right now every door is closed until we look.

Here is why it sits first. The test reads several genetic features at once, and roughly four in ten people with this cancer turn out to have at least one feature that a targeted drug can act on. The test itself does not treat the cancer. What it does is tell you which, if any, of the options further down this page are even reachable.

It is also about as low-risk as a test gets. Usually it can be run on a tumor sample that was already taken and stored, so no new procedure is needed; when stored tissue is not enough, a blood draw can sometimes stand in. Results generally come back in one to three weeks. There is no toxicity to weigh against, which is part of why it was an easy call.

One practical note worth raising with your team: a few features need their own specific tests rather than the standard panel. A marker called HER2 is best checked by staining the tumor tissue directly, and a feature called mismatch-repair deficiency is fastest to confirm with a tissue stain too. And catching an FGFR2 fusion reliably takes a test that reads RNA, not DNA alone. So the right order is the full genetic panel, run with an RNA arm, plus those couple of tissue stains, all together.

The options

Everything in this section depends on the test above coming back positive for the matching feature. None of these is something you start tomorrow; each one waits on a specific result. They are listed in the order Libby ranked them, which reflects a mix of how strong the evidence is and how likely the feature is to show up in this cancer. Two things hold true throughout: most of these options ride on a test result that is not yet in hand, and some of the newer drugs are still early or trial-stage. I will flag which is which as we go.

Option 1 — Ivosidenib (only if the tumor has an IDH1 R132 change)

This option is only available if the test comes back positive for a genetic change called IDH1 R132. If it comes back negative, this option is off the table, and Libby's recommendations on this page are limited to treatments aimed at a specific genetic feature, so a negative result here does not by itself leave you with nothing; it just means this particular door is closed and you move on to the others. (More on what happens if every door is closed at the end of this section.)

Ivosidenib is a once-a-day pill. Of all the options on this page, it rests on the strongest kind of evidence: a trial where people were randomly assigned to the drug or a placebo, which is the most reliable way to know a drug is doing something. In that trial, the drug roughly doubled the time before the cancer grew, from about 1.4 months to about 2.7 months on average, and that difference was statistically convincing.

Honesty matters here. The tumors very rarely shrank on this drug, only about 2 in 100 had measurable shrinkage. The realistic goal is to hold the cancer steady for a while, not to make it smaller. On whether it helps people live longer, the published numbers point toward a benefit, but the cleanest version of that analysis did not quite clear the bar for certainty, and the more favorable number leans on a statistical adjustment made after most placebo patients later crossed over to the drug. So I can't give you a precise survival improvement here without overstating what the trial actually showed.

Side effects were mild overall, close to placebo. The one thing that needs watching is an effect on heart rhythm, checked with an ECG before and during treatment. This change shows up in maybe 15 to 20 percent of intrahepatic (inside-the-liver) cholangiocarcinomas, so it is one of the more likely features to turn up.

Option 2 — Futibatinib (only if the tumor has an FGFR2 fusion)

This option is only available if the test comes back positive for an FGFR2 fusion or rearrangement. If it comes back negative, this option is off the table.

Futibatinib is a daily pill, and the FGFR2 fusion it targets is the single most common targetable change in this cancer, roughly 1 in 8 of the inside-the-liver cases. In its trial, about 42 percent of tumors shrank, and people lived a little under two years on average. Very few had to stop the drug because of side effects. What gives this option extra weight is that a second, separate drug in the same family showed a similar result, which is the closest thing on this page to one trial confirming another. This drug also tends to keep working against some of the resistance the cancer develops to the gentler relative below it, which matters if you need to switch later.

The tradeoff is a side effect called high blood phosphate. It is a built-in consequence of how the drug works, and it reaches a more serious level in roughly 30 percent of people, so it needs active management with diet and medication. The gentler relative of this drug is the next option, and the choice between the two comes down to this: futibatinib has the slightly stronger response data and may still work after the gentler one stops, while the gentler one is easier on the body. That is a genuine conversation to have, not a settled answer.

The fine print: the evidence is from a single-group study, everyone got the drug with no comparison group, and the response numbers are a stand-in for the harder question of how much longer people live. The trial was also built around inside-the-liver cases, so if the cancer started outside the liver the fit is weaker.

Option 3 — Pemigatinib (also only if the tumor has an FGFR2 fusion)

This option is only available if the test comes back positive for an FGFR2 fusion. It targets the exact same feature as the option above, so the two are alternatives within the same lane rather than separate paths.

Pemigatinib is a pill taken on a schedule of two weeks on, one week off. About 36 percent of tumors shrank in its trial, in line with futibatinib. It was the first drug of its kind approved for this cancer, so it has the longest real-world track record. Its version of the high-phosphate side effect tends to stay milder, common but rarely at the serious level, which is the whole argument for choosing it over futibatinib when going gentler matters more. It does carry one effect worth knowing about: a fluid buildup behind the retina that needs scheduled eye checks.

Same fine print as above: single-group study, response numbers standing in for survival, and the reported survival figure is too early to lean on.

Option 4 — Lirafugratinib (also only if the tumor has an FGFR2 fusion)

This option is only available if the test comes back positive for an FGFR2 fusion. It targets the same feature as the two options above. **This option carries caveats.** Here is the tradeoff to weigh.

Lirafugratinib is a newer pill designed to hit FGFR2 more selectively. In an early study, its response numbers were the strongest in this lane, about 47 percent of tumors shrank, and because of its more selective design it largely skips the high-phosphate side effect that the two approved drugs cause. On paper that reads like the best of both worlds: higher response and an easier phosphate profile.

The caveats are what keep it here rather than at the front of the lane, and they are worth stating plainly. First, this drug is not yet approved. The only ways to get it in the US right now are through a clinical trial slot or an early-access route while its approval decision is pending. Second, the response figure comes from a conference presentation, not a fully published, peer-reviewed paper, so the underlying details cannot be checked as thoroughly as the numbers for the two approved drugs above. Third, the gentler phosphate profile is not a free win: this drug brings its own tough side effect, a severe hand-and-foot skin reaction that reaches a serious, function-limiting level in about 1 in 3 people. So swapping one drug's side effect for another's is closer to a lateral move than a clear upgrade. If you are drawn to trials, this is a real and higher-response door worth asking about, with those three caveats fully on the table.

Option 5 — Zanidatamab (only if the tumor is HER2-positive)

This option is only available if the test comes back positive for HER2. If HER2 is negative, this option is off the table. Worth noting: HER2 has to be confirmed by staining the tumor tissue, not by the DNA panel alone, and it tends to show up more in cancers that started outside the liver.

Zanidatamab is an infusion given every two weeks. The reason it leads the HER2 options is that its trial was done specifically in bile-duct cancer, rather than borrowing results from a grab-bag of different cancers. In that trial, about 41 percent of tumors shrank, and responses lasted a meaningful stretch, around a year on average for those who responded.

The main thing to watch is the heart's pumping strength, monitored with periodic echocardiograms; a more serious drop happened in about 3 percent of people. Some people also had infusion reactions or diarrhea. The evidence is again from a single-group study, so it carries that limitation. There is a second HER2-targeting drug that was considered but held below this one because of a lung-inflammation risk; that one is described further down.

Option 6 — Dabrafenib plus trametinib (only if the tumor has a BRAF V600E change)

This option is only available if the test comes back positive for a specific BRAF change called V600E. If that exact change is not present, this option is off the table. The "exact" part matters: a quick screening stain hinting at BRAF is not enough, and other BRAF changes do not qualify, so it has to be confirmed precisely.

This is a pair of pills taken together. In its trial, about 47 percent of tumors shrank, a strong result, and the cancer was held in check for around 9 months on average. A confirmed V600E change gives two separate routes to access these drugs, so getting the drug is usually not the hard part.

The reason this sits below the gentler pill options, despite the strong response, is the side-effect load. Recurring fever is a characteristic effect of this combination, and serious side effects of some kind happened in roughly 40 percent of people. None of the trial deaths were tied to the drugs, and there is a well-worn playbook for managing the fever, but it is a heavier regimen day to day. This change is uncommon in this cancer, somewhere in the 1 to 5 percent range, and the trial group was small.

A HER2 option that was considered but held back — trastuzumab deruxtecan

There is a second HER2-targeting drug, trastuzumab deruxtecan, that was looked at and **carries serious caveats** , which is why it sits below zanidatamab rather than beside it. It is on the page because Libby shows what was weighed and where it pulled back, not only what it kept. Like zanidatamab, it only applies if HER2 comes back positive.

In the group whose tumors most strongly expressed HER2, its response numbers were competitive with zanidatamab. The concern is a specific lung inflammation that showed up in about 1 in 10 people in its trial, with three treatment-related deaths, and there was no baseline lung assessment on file to judge that risk against. With zanidatamab covering the HER2 situation on bile-duct-specific data and a lighter safety profile, this drug was held in reserve. It could come back into the conversation, but only with a confirmed HER2 result, a baseline lung check, and a written monitoring plan in place first. One more note on the biomarker: the approval here is for tumors that express HER2 at the strongest level. If the tumor is only "HER2-low," that is a weaker signal that this drug will help, and using it there would be experimental rather than standard, so a low result should not be read as the same green light as a strong one.

Option 7 — Pembrolizumab (only if the tumor is MSI-high / mismatch-repair deficient or has high TMB)

This option is only available if the test comes back positive for one of two related immune-system markers : mismatch-repair deficiency (also called MSI-high) or a high tumor mutational burden (TMB). If neither shows up, this option is off the table. These markers are uncommon in this cancer, roughly 1 to 3 percent, but they change the picture completely when present.

Pembrolizumab is an immunotherapy infusion given every three weeks. It works by helping the immune system recognize the cancer, and when it works, the benefit can be unusually durable. In the relevant studies, about 29 to 34 percent of tumors shrank depending on which marker qualified the patient, and people lived close to two years on average.

Two honest limits. First, these numbers come from studies spanning many cancer types with no bile-duct-specific group, so the size of the benefit here is inferred rather than measured. Second, immunotherapy can push the immune system into attacking healthy organs; serious immune-related side effects happened in about 15 percent of people, and there was one treatment-related death in the study. These effects have an established management approach, but they are real and need close monitoring. One more note: the high-TMB marker is a less well-proven predictor in this cancer than mismatch-repair deficiency, so a positive there is a softer signal than an MSI-high result.

What if the test comes back negative for everything

This is the case worth naming plainly. If the genetic test finds none of these features, then none of the options on this page applies, because every one of them is keyed to a specific feature being present. Libby's recommendations here are limited to feature-targeting treatments, so a fully negative test means there is nothing further for Libby to rank.

That does not mean there is no treatment. Standard care for metastatic cholangiocarcinoma exists and is the path most people take. It is simply outside the scope of this page, and it is a conversation to have with your oncologist, who knows your full situation. I am deliberately not naming specific standard-care drugs here, because choosing among them is your treating team's call, not Libby's.

One more thing: these features are checked independently. A negative result on one does not rule out the others. A tumor could be negative for the IDH1 change but positive for an FGFR2 fusion, and the FGFR2 options would still be live.

Questions to ask your oncologist

- Has my tumor had comprehensive genomic profiling yet, and if not, can we order it now using stored tissue or a blood test rather than a new biopsy?
- Will the testing include the DNA panel, an RNA arm for FGFR2 fusions, and the separate tissue stains for HER2 and mismatch-repair, so we don't miss anything that needs its own test?
- Did my cancer start inside the liver or outside it, and how does that change which of these genetic features is most likely?
- If the genetic test comes back negative for all of these features, what standard treatment would you recommend, and where do clinical trials fit in?
- For whichever targeted option my result points to, what is the plan for monitoring its main side effect (the heart-rhythm checks for ivosidenib, the phosphate management for the FGFR2 drugs, the echocardiograms for zanidatamab, the fever for the BRAF pills, or the lung monitoring for the immunotherapy)?
- If I turn out to have an FGFR2 fusion, how would we choose among futibatinib, pemigatinib, and the trial drug lirafugratinib for me specifically, given the differences in side effects and approval status?
- If HER2 is positive, why zanidatamab over trastuzumab deruxtecan for me, and what lung monitoring would we need if the second drug ever came up?
- Before starting any of these, do we have the baseline tests we'd need (an ECG, an echocardiogram, lung function), and should we get them now so we're ready?

Sources

The recommendations on this page draw on the following published trials, conference reports, and registries.

PubMed (PMID): 31682550, 32203698, 32416072, 32818466, 32919526, 34848557, 36652354, 36914633, 37276871, 37870536.

Conference report (DOI): 10.1200/JCO.2026.44.2_suppl.476.

ClinicalTrials.gov (NCT): NCT02034110, NCT02052778, NCT02628067, NCT02924376, NCT02989857, NCT04466891, NCT04482309, NCT04526106.